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Differences of the stem vascular system across populations of two tropical species under contrasting water conditions

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ABSTRACT

Structural differences in the secondary vascular tissues among habitats can contribute to understanding species performances, especially regarding water and photosynthate transport. The pattern of association between the secondary xylem tissue and water availability from the environment has been widely studied, unlike the secondary phloem, which has been barely explored. Here, we evaluated the structural variation of the secondary xylem and phloem in stems of four populations of two tropical tree species under contrasting water conditions. We also investigated the mirrored structure between both tissues. At dry sites, *Moquiniastrum polymorphum* had higher vessel density, thicker xylem fibers cell walls, and taller rays in both tissues commonly associated with safe transport, in agreement with our expectations. In contrast, the populations of *Zanthoxylum rhoifolium* had most features in disagreement with the water availability of each site. The perforation and sieve plates, the ray composition, and the axial parenchyma were similar in the two tree species' xylem and phloem tissues. However, the quantitative descriptors of cell sizes were not correlated between the xylem and phloem. In general, there is a different pattern of morphological variation across sites in the two tropical tree species, highlighting that any generalization regarding the vascular system structure across environments should be avoided. Xylem and phloem revealed a mirrored structure in a few qualitative features, not followed by the dimensions of different cell types. Future research needs to explore the causes of the unexpected structural variation in the vascular system across populations in tropical tree species.

Keywords: Asteraceae; Atlantic rain forest; cerrado; *Moquiniastrum polymorphum*; Rutaceae; secondary phloem; secondary xylem; *Zanthoxylum rhoifolium*.

INTRODUCTION

The production of the secondary vascular system by the cambium gives rise to most of the lateral growth of woody plants. The secondary vascular system flows resources through the plant body. Water and mineral nutrients are carried through the xylem, whereas sugars, amino acids, micronutrients, lipids, hormones, and RNAs are translocated through the phloem (Lucas *et al.* 2013). Also, the vascular structure provides mechanical support to the plant body, mostly attributed to the secondary xylem (Pratt & Jacobsen 2017), while the secondary phloem can be involved in posture control (Clair *et al.* 2019). Finally, the secondary vascular system can function as water and sugar storage (Evert 2006; Pratt & Jacobsen 2017). These three functions of the vascular tissue (transport, mechanical support, and storage) are assigned to specialized cells: conducting elements, the sclerenchyma, and the parenchyma, respectively (Evert 2006). The structural and functional differences of the vascular system across individuals and populations can be explained by changes in environmental conditions (Sultan 2000). For example, under drier conditions, the xylem generally shows vessel features associated with safer water transport (Hacke *et al.* 2017; Tng *et al.* 2018) and fibers and parenchyma features related to drought resistance (Tng *et al.* 2018; Janssen *et al.* 2020). For the secondary phloem, drought might impact phloem transport, although the relations between phloem transport during the drought and anatomy are unclear (Sevanto 2018). Therefore, structural adjustments in the vascular system across sites

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are essential to sustain water and sugar supply for plant development, promoting or inhibiting plant growth and, ultimately, its survival (Qaderi *et al.* 2019).

The secondary xylem of stems has been widely studied, and its structural differences, at the level of tissue and cells, are closely associated with water availability. Under permanently or seasonally dry conditions, oak species have produced secondary xylem with reduced vessel area and a higher amount of axial parenchyma (Fonti *et al.* 2013). A lower vessel area, which is strongly related to vessel diameter reduction, may indicate safe water transport, as narrow vessels are more resistant to cavitation (Sperry *et al.* 2008). A larger area occupied by the parenchyma can confer drought resistance through its higher water storage (Borchert & Pockman 2005, Słupianek *et al.* 2021). It is also worth mentioning that parenchyma plays an active role in refilling embolism vessels daily (Bucci *et al.* 2003). The ground tissue, composed of a fraction of fibers, can improve the drought resistance as a higher area of fiber lumen increases wood capacitance (Pratt & Jacobsen 2017; Janssen *et al.* 2020). Also, a higher fiber wall area increases the tissue mechanical strength, preventing implosions resulting from the high vessel negative pressure caused by drought (Jacoben *et al.* 2005). Thus, the variation in water availability in the environment can explain trade-offs among fractions of the xylem cells across plants and populations.

At the cellular level of the secondary xylem, the number, size, and clustering of the conducting cells can be associated with the local environmental conditions. Narrow, short, more numerous, and clustered vessels are features generally found in the xylem of plants from sites with lower water availability (Carlquist 1966; Baas *et al.* 1983; Alves & Angyalossy-Alfonso 2000; Wheeler *et al.* 2007; Pfautsch *et al.* 2016; Hacke *et al.* 2017; Frankiewicz *et al.* 2021). These cellular features would increase protection against cavitation and embolism (Hacke *et al.* 2017), maintaining vessel integrity under high tension (Sperry *et al.* 2006), and the water flows through alternative routes (Tyree *et al.* 1994). On the other hand, long and wide vessel elements have been associated with plants in higher water availability sites. In this case, larger vessels transport a higher water volume per unit area and reduce the resistance to the water flow by a higher distance between the terminal walls, increasing the efficiency of xylem water transport (Sperry *et al.* 2008). Pits play a primary role in water transport through the xylem, in which the intervessel pits reduce the resistance for hydraulic connectivity between neighboring vessels (Carlquist 2001), and the vessel-ray pits are effective in the radial transport of osmotically active sugar (Bucci *et al.* 2003). However, pit sizes were not related to the water availability across habitats (Sonsin *et al.* 2012, Longui *et al.* 2014).

The relationship between the sizes and anatomical features of the xylem parenchyma and fibers with environmental water availability is much less clear (Wheeler *et al.* 2007; Morris *et al.* 2016). For example, from permanently or seasonally dry regions, plant species commonly have thick-walled xylem fibers (Alves & Angyalossy-Alfonso 2002; Sonsin *et al.* 2012; Warwick *et al.* 2017) that increase mechanical support for the conducting elements under high negative pressure, preventing them from collapsing (Hacke *et al.* 2001; Pratt & Jacobsen 2017). In addition, fibers with larger lumens may increase the wood capacitance (Pratt *et al.* 2007) since they increase the space for water that can be transferred to the vessels when the water demand exceeds the supply (Pratt & Jacobsen 2017). So, secondary xylem with fibers with thicker walls and large lumens can provide drought resistance to plants in dry sites (Janssen *et al.* 2020). Further, plant species with wider, taller, and numerous rays and more abundant axial parenchyma (Alves & Angyalossy-Alfonso 2002; Sonsin *et al.* 2012; Warwick *et al.* 2017) would have a higher hydraulic capacity of the xylem (Morris *et al.* 2016, 2018) and drought resistance in drier regions (Janssen *et al.* 2020), given the facility for storage and transport of water over short distances (Evert 2006), especially during periods of plant dehydration (Borchert & Pockman 2005).

In contrast to the secondary xylem of stems, the structural differences in the secondary phloem according to the water availability in the environment have been barely explored at the tissue level. The variation of the fraction of cells with primary walls, that is the sieve elements and parenchyma, and sclerenchyma with secondary lignified walls across plant populations is unknown according to our revision until now. At the level of the cells, the pattern of variation of the shape and dimensions of the conducting elements is unclear, as wider sieve tubes were observed in sites of both high (Vergílio & Marcati 2017) and low water availability (Sevanto *et al.* 2018; Kiorapostolou & Petit 2019), while narrower sieve tubes were found in species under drought regimes (Salmon *et al.* 2019). Under drier conditions, the sieve element integrity can be maintained through the parenchyma and the sclerenchyma surrounding the conducting cells (Sevanto 2018). However, thicker cell walls of the sclerenchyma around the sieve elements were found in *Stryphnodendron polyphyllum* (Leguminosae) in a wet site, providing support for the inner bark in this species (Vergílio & Marcati 2017). Thus, although parenchyma and sclerenchyma may be fundamental in maintaining water supply and mechanical strength favoring plant survival during drought (Sevanto 2018), a pattern of variation of these cells is unclear according to the environmental conditions. Notably, there is a correspondence between the variation of the ray dimensions and the water availability, with wider (den Outer 1993) and taller (Vergílio & Marcati 2017) phloem rays occurring in plants in drier environments. Wider and taller rays could contribute to enlarging the storage capacity and transport of water across the vascular system (Hölttä *et al.* 2006), especially under dry conditions (van Bel 1990; Sokolowska & Zagórska-Marek 2012).

Apparently, a pattern of variation of the xylem structure of stems in wet and dry sites is more notable than the phloem, which may result from a lack of detailed studies that compare the secondary phloem structure among plant populations and habitats. Indeed, new evidence indicates that the phloem can play a crucial role in plant performance under drought conditions since parenchyma and conducting cells are a source of water supply to the xylem that could reduce their negative water pressures (Savage *et al.* 2016). Furthermore, long-distance water and photosynthate transport operate in a water potential equilibrium between the xylem and phloem (Thompson & Holbrook 2003a; Hölttä *et al.* 2006). This equilibrium can be sustained by a functional coupling between xylem and phloem, represented by similar structural features (Kiorapostolou & Petit 2019), such as congruent diameters, length, and density of the conducting cells; similar size and the number of the rays; and fibers with the same size and with their walls similarly thickened. These corresponding features can characterize a mirrored structure between the xylem and phloem into the stems.

In this study, we explored the variation of the secondary vascular system in stems of two widely distributed tree species in the Neotropics. We chose four populations of these two co-occurring species across habitats to investigate the pattern of cross-population variation in the secondary xylem and phloem. Specifically, we had two main questions: (i) How do the xylem and phloem tissues of the stem vary across populations of these two tree species under different water availability conditions? And (ii) is there a mirrored structure of the conducting, support, and reserve cells between the xylem and phloem tissues in each tree species, regardless of environmental conditions?

For the first question, we expected that the xylem and phloem structure of stems would differ across populations in both species. In-plant populations under drier conditions, we expected xylem tissues with a lower fraction of vessels and higher fractions of fibers and parenchyma (axial and radial). We also expected more numerous and closely grouped vessels at the cell level, with reduced diameters and intervessel and vessel-ray pits, shorter vessel elements, fibers with wider lumens and thicker walls, and larger, taller, and numerous rays. All these features contribute to safe water transport in drier sites. In contrast, we expected an opposite pattern of xylem features in a plant population growing in wetter conditions, favoring a more efficient water transport. Regarding the phloem, we expected a higher proportion of cells with primary walls (sieve tubes and parenchyma) and lower fractions of sclerenchyma with secondary lignified walls in populations under drier conditions of both species. We also expected narrower sieve tubes at the cell level, larger, taller, and numerous rays, and thicker secondary lignified walls in plant populations growing in drier conditions. All these features could aid in plant resistance to droughts. In contrast, in a plant population growing under wetter conditions, we expected an opposite pattern of phloem features.

For the second question, we expected the anatomical features of the xylem and phloem would have a mirrored structure in both plant species. In this case, we would find a correlational pattern of diameters, length, and density of the vessels and sieve tubes between tissues; similar width, height, and number of rays; and correlated diameters, length, and wall thickening of the sclerenchyma. Also, we expected similar types of perforation and sieve plates, a similar arrangement and distribution of the axial parenchyma, and the same cellular composition of the rays in both xylem and phloem across populations of each tree species.

MATERIALS AND METHODS

Plant species, study areas, and experimental design

We selected two semi-deciduous and phylogenetically distant tree species: *Moquiniastrum polymorphum* (Less.) G. Sancho (Asteraceae; asterids) and *Zanthoxylum rhoifolium* Lam. (Rutaceae; rosids). *Z. rhoifolium* is widely distributed in the Neotropics, from Central America until the central and southern boundary of South America (Pirani & Groppo 2020). *M. polymorphum* is well distributed throughout the Central South America zone, from the interior of the continent to the Atlantic coast (Sancho & Roque 2020). Both species are distributed in regions with a seasonal climate but with different environmental conditions across sites, mainly in relation to the distribution of precipitation throughout the year (seasonality of precipitation) and the physical soil structure, both factors critically determine the soil water availability for plants.

To test our hypothesis as stated in the introduction, we first selected four sites in the State of São Paulo, Brazil (Fig. 1A): one site was classified as the wet site, and three additional sites were classified as dry sites. Our classification was based on the intersection between the precipitation pattern over the year, especially in the dry season, and the physical structure of the soil (Table 1; Appendices A and B at 10.6084/m9.figshare.16903063). The wettest site (called 'wet site' from now on) was chosen in the Parque Estadual Carlos Botelho, São Miguel Arcanjo municipality (24°00′ to 24°15′S and 47°55′′ to 48°05′W), approximately 90 km from the coast (Fig. 1A). The three remaining sites were chosen farther from the coast. The "dry1" site is located 230 km from the coast, at the Fazenda Palmeira da Serra, Pratânia municipality (22°48′ to 22°49′S and 48°45′ to 48°44′W; Fig. 1A). The "dry2" site is 315 km from the coast, at the Caetetus Ecological Station, Gália municipality (22°41′ to 22°46′S and 49°10′ to 49°16′W; Fig. 1A). Finally, the "dry3" site is 360 km from the coast, at the Assis Ecological Station, Assis municipality (22°33′ to 22°38′S and 50°21′ to 50°24′W; Fig. 1A).

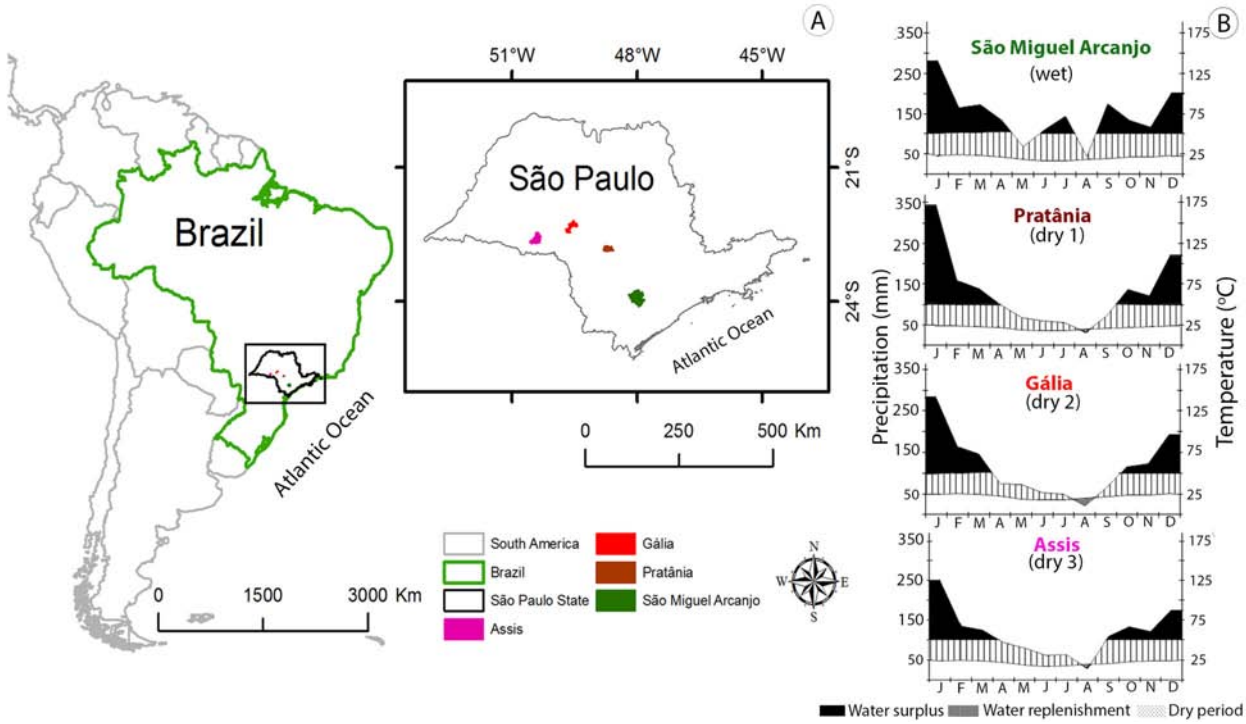


Figure 1. Map with (A) the geographic location of each site, and (B) the climatic diagrams describing the monthly variation in precipitation and mean temperature.

Table 1.
Vegetation types and climatic and edaphic descriptors of the study sites.

	Wet	Dry1	Dry2	Dry3	Graphs
Total annual rainfall (mm) [a]	1539	1403	1286	1308	Appendix B (Fig. B1A)
Total rainfall of the wet season (mm) [a]	948	1012	915	873	Appendix B (Fig. B1B)
Total rainfall of the dry season (mm) [a]	591	390	371	419	Appendix B (Fig. B1C)
Seasonality Index	0.52	0.61	0.61	0.6	Appendix B (Fig. B1D)
Relative humidity (%) [b]	78.5	59.9	69	68.7	–
Sand (g kg^{-1}) [c]	390	870	860	810	–
Clay (g kg^{-1}) [c]	500	99	120	155	–
Silt (g kg^{-1}) [c]	110	31	20	35	–
Soil moisture (%) [c]	39	8	12	13	–

[a] Provided by the Meteorological Station of the Faculdade de Ciências Agronômicas, UNESP, Botucatu (dry1); the Integrated Agrometeorological Information Center (http://www.ciiagro.sp.gov.br) (dry2 and dry3); and the Agrometeorological Monitoring System (https://www.agritempo.gov.br/agritempo) (wet).

[b] Provided by the Meteorological Station of the Faculdade de Ciências Agronômicas, UNESP, Botucatu (dry1) and from Destefani *et al.* (2006) (wet, dry2 and dry3).

[c] Obtained from Cooper *et al.* (2013) (wet); Lara et al (2017) (dry1); Cooper *et al.* (2012) (dry2); Juhász *et al.* (2006) (dry3).

The climatic diagrams of each site (Fig. 1B) show that in the dry period, the wet site has only two non-consecutive months of water deficit (<100 mm of precipitation) in May and August. The other three dry sites have an average of 5 consecutive months of water deficit (<100 mm of precipitation) from April/May to August/September, distinguishing the dry season. Considering the accumulated rainfall in the dry season (April to September), the wet site received, on average, 200 mm more rainfall than the other dry sites (Table 1; Appendix B at 10.6084/m9.figshare.16903063). This difference is highlighted by the seasonality index (SI) showing a less seasonal pattern for the wet site (Table 1; Appendix B at 10.6084/m9.figshare.16903063). Additionally, the wet site has a more clay-type soil that favors water retention especially in the dry period, which can be also evidenced by the soil moisture pattern (Table 1), contrary to what happens in the sandy soil of the three dry sites. Together the different descriptors of soil and water availability allowed us to classify the sites as proposed above, only one wet site near

the coast, and three dry sites much further away from the coast. Therefore, we expect that the wet site with clay-type soils and more intense precipitation in the dry season generate a smaller water deficit for plants, opposite to the three dry sites with sandy soil and more scarce precipitation in the dry season.

For more details of vegetation, climate, and soil descriptors of each site see Appendix A at 10.6084/m9.figshare.16903063.

Collection of the vascular system

We selected one population of five mature trees in each site. In total, 20 individuals of each tree species were sampled (Appendix C at 10.6084/m9.figshare.16903063). In each plant, we obtained samples of the secondary xylem and bark samples non-destructively, using a hammer and chisel, from the main stem at 1.3 m above ground, between March and June 2014. Each sample of the vascular system measured about 4 cm wide × 4 cm long and 5 cm deep.

In each sample, we first separated the secondary xylem from the bark. The xylem samples were divided into five subsamples of approximately 1 cm^3, which were fixed and kept in FAA 70 for five days, and then stored in 70% ethanol. Histological sections (17 μm) were made using a sliding microtome (Reichert Om E, Vienna, Austria) on the transverse, tangential, and radial faces; and were stained with both aqueous 1% safranin and 1% Astra blue (1:5) (Bukatsch 1972). We dehydrated the sections in an increasing ethanol series and mounted them in synthetic resin on permanent slides.

The bark samples were divided into six subsamples of approximately 1 cm^3, which were fixed in glutaraldehyde and formaldehyde (Karnovsky 1965, modified by Kraus & Arduin 1997) for 40 h and then stored in 70% ethanol. The subsamples were embedded in polyethylene glycol 1500 (Barbosa *et al.* 2010), and transverse, tangential, and radial sections were made using a Reichert Om E sliding microtome (13–15 μm). During the sectioning, we used plastic adhesive on the samples to avoid histological damage (Barbosa *et al.* 2010). We stained the histological sections with a solution of 0.1% ethanolic basic fuchsin and 1% aqueous Astra blue (1:5) (Bukatsch 1972, replacing safranin with basic fuchsin), or a solution of 2% aqueous acridine orange and 1% Astra blue (1: 5) (Bukatsch 1972, replacing safranin with acridine orange). We used adhesive gelatin (Haupt 1930) to attach the sections to the slides, and synthetic resin to assemble permanent slides.

To make more detailed observations of the phloem, we took a subsample of bark from each plant and divided it into at least two smaller subsamples of approximately 0.5 cm^3, dehydrated them in an ethanol series, and embedded them in plastic resin (Bennett *et al.* 1976). We made transverse and longitudinal radial sections 4–5 μm thick, using a semi-automated rotary microtome (Leica RM2245; Leica, Buffalo Grove, IL, USA). We stained the sections with aqueous toluidine blue (0.1%) in acetate buffer, pH 4.7 (O'Brien *et al.* 1964). We mounted temporary slides in water at the time of the analysis.

From samples of the sapwood and the conducting phloem, we dissociated the cellular elements of the xylem and phloem in glacial acetic acid and hydrogen peroxide (1:1), respectively, in an oven at 60°C (Franklin 1945, modified by Kraus & Arduin 1997). The periods required to dissociate the secondary xylem and phloem were 24 h and about 5 h, respectively. We stained the dissociated secondary xylem with ethanolic safranin 1% and the secondary phloem with aqueous Astra blue 1% and assembled semi-permanent slides in 50% glycerol.

Anatomical descriptors of secondary xylem

For each plant, we assembled 10 permanent histological slides containing transverse, tangential, and radial sections, and 20 semi-permanent slides containing the dissociated material. We obtained images of each slide, using a Zeiss Axiocam MRc camera (Zeiss, Göttingen, Germany) coupled to a Zeiss Axioskop 40 microscope. Fiber, vessel, and parenchyma (axial and radial together) fractions were estimated in the sapwood, in cross-section, in 15 known areas for each plant at each site, using ImageJ version 1.6.0 (https://imagej.nih.gov/ij/).

For the qualitative description of secondary xylem, a total of 40 qualitative descriptors were used to classify vessels, fibers, and parenchyma in different character states, following the recommendations of the IAWA Committee (1989).

We assessed 15 quantitative descriptors in the sapwood, totaling 30 measurements of each anatomical descriptor per plant. The anatomical descriptors were: vessel density (n mm^{-2}); vessel diameter; length of vessel elements (μm); diameter and aperture of intervascular and vessel-ray pits (μm); fiber diameter, fiber lumen diameter, fiber wall thickness, and fiber length (μm); number of rays per millimeter (n mm^{-1}); and height and width of the rays (μm). For each xylem descriptor, we followed the recommendations for measurement (analysis and the use of histological sections or dissociated material for measurements) of the IAWA Committee (1989). All measurements were performed with the Zeiss AxioVision 4.9.1.0 image analysis system. We also estimated the vessel grouping index, first selecting randomly 25 groups of vessels, including both solitary and multiple vessels in each group; and second, counting the number of vessels in each group in the cross-sections. We then summed the number of vessels in all groups and divided by 25 to obtain the mean number of vessels per group. A value of "1" indicates exclusively solitary vessels, and as the value of the index increases, the greater the degree of grouping (Carlquist 2001).

Anatomical descriptors of secondary phloem

For each plant, we assembled 13 permanent histological slides of the bark and 20 semi-permanent slides containing the dissociated phloem conducting cells. Images of each slide were obtained as for the xylem. We selected one area containing the phloem cells with primary walls and secondary lignified walls in cross-sections. We measured the fraction of cells with secondary lignified walls, which was subtracted from the total area, thus obtaining the fraction of cells with primary walls.

For the qualitative description of secondary phloem, 22 qualitative descriptors were used to classify sieve tubes, sclerenchyma, and parenchyma into different character states, following Angyalossy *et al.* (2016).

We measured five and nine phloem quantitative descriptors in *M. polymorphum* and *Z. rhoifolium* respectively, totaling 30 measurements of each anatomical descriptor per plant. The descriptors shared between both species were: the diameter of sieve tubes (μm), length of sieve-tube elements (μm), number of rays per millimeter (n mm^{-1}), and height and width of the rays (μm). In the secondary phloem of *Z. rhoifolium*, we also measured, in the fibers, the total diameter, lumen diameter, wall thickness, and length (μm) of cells. In the secondary phloem of *M. polymorphum*, fibers were absent, and no sclerenchyma was measured. For the measurements of each phloem descriptor, we followed the recommendations of Angyalossy *et al.* (2016).

Statistical analyses

Statistical analyses to evaluate the cross-population variation of the secondary vascular system were performed separately for each species. To evaluate whether there were differences in the quantitative anatomical descriptors of secondary xylem and phloem across the four populations, we first performed a principal components analysis (PCA) to reduce the anatomical descriptors in a few orthogonal axes. For the PCA, we used the mean value of each quantitative anatomical descriptor for each plant, totaling 20 values of each descriptor for each tree species (five plants per population). Subsequently, we tested the correlation of all pairs of descriptors, using Pearson's correlations with the Holm adjustment method (Holm 1979). When we found a significant correlation between a pair of descriptors, only one was retained in the PCA analysis. The criterion to keep one of the highly correlated descriptors was empirical studies (Carlquist 1966; Baas *et al.* 1983; Alves & Angyalossy-Alfonso 2000; 2002; Wheeler *et al.* 2007; Sperry *et al.* 2008; Sonsin *et al.* 2012; Longui *et al.* 2014; Pfautsch *et al.* 2016; Hacke *et al.* 2017) supporting the hypothesis of a functional relationship between these anatomical descriptors and the water availability in the environment. The reduction in the total number of anatomical descriptors used in the PCA analysis was also an advantage here since the number of descriptors should be lower than the sample size (Legendre & Legendre 2012). We eventually used 17 quantitative descriptors for *M. polymorphum* and 16 for *Z. rhoifolium* in our PCA analyses, lower than the sample size of 20 plants per species. The quantitative descriptors that remained in the PCA analysis in both species were: diameter, density, and vessel grouping index; length of vessel elements; length and wall thickness of xylem fibers; number, height, and width of xylem rays; vessel-ray pit aperture; xylem parenchyma fraction; sieve-tube diameter; sieve-tube element length; phloem ray height; and the fraction of cells with primary walls in the phloem. Additionally, the intervessel pit aperture and the number of rays per millimeter in the phloem were selected only for *M. polymorphum*, and the thickness of the phloem-fiber wall was selected only for *Z. rhoifolium*. These quantitative descriptors were standardized, and the PCA analysis was performed. Distinct eigenvalues (λ) of the axes are a fundamental assumption of PCA, and they need to be larger than expected by random data to be useful in the axes selection and indicative of biological meaning (Björklund 2019). For that, we adopted both Kaiser–Guttman and broken stick model criteria (Borcard *et al.* 2011; for details, see Appendices D and E at 10.6084/m9.figshare.16903063) to evaluate the PCA performance. The common axes retained by the two criteria were selected to represent the amount of variance explained. Thus, for *M. polymorphum*, we considered the first three axes (Appendix D at 10.6084/m9.figshare.16903063), and for *Z. rhoifolium* the first four axes (Appendix E) at 10.6084/m9.figshare.16903063.

The axes of the PCA representing the anatomical features were used as response variables, and the populations under different environmental conditions were used as fixed categorical factors in general linear models (GLM) with a Gaussian probabilistic distribution. We also included the tree height as a continuous fixed factor in the models since the tree height is considered a good predictor of cell size variation of the xylem (Olson *et al.* 2018; Fajardo *et al.* 2020) and phloem (Clerx *et al.* 2020) tissues. The inclusion of tree height in statistical models became necessary to control any scaling relationship between the cell sizes or tissue amount and the plant size as a whole, as both increase during plant development.

To evaluate the mirrored structural anatomy between the secondary xylem and phloem, we reduced the quantitative anatomical descriptors performing two principal components analyses (PCA) separately, one for xylem and another for phloem. The first two axes of each PCA were used in simple linear regressions, the first two phloem axes being the response variables, and the first two xylem axes and the tree height the explanatory variables. We considered the models' higher adjusted R^2 as indicative of the mirrored structure between the secondary xylem and phloem quantitative descriptors.

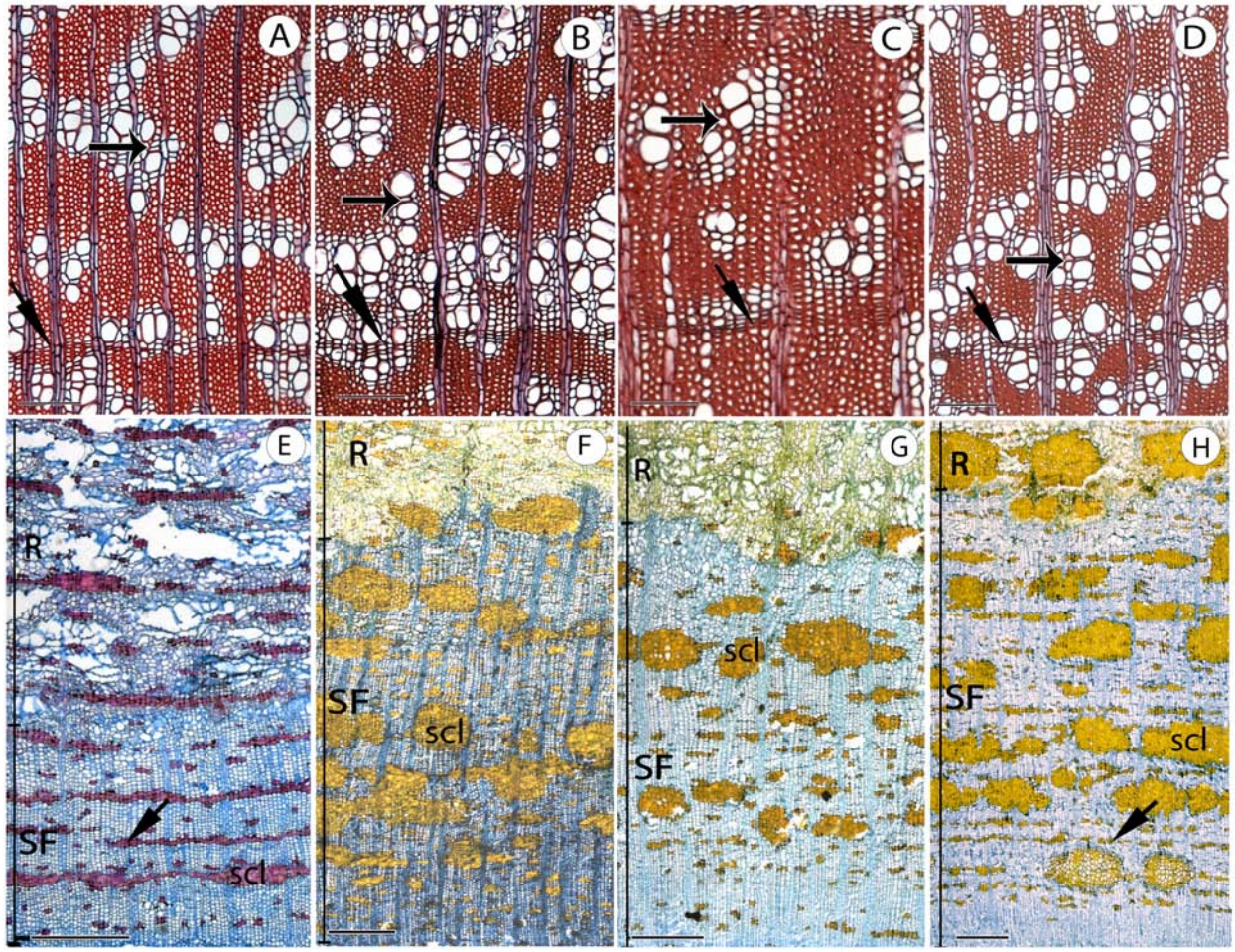


Figure 2. Cross-population variation of the secondary vascular system anatomy in the stems of *Moquiniastrum polymorphum* in transverse section (A, E, wet; B, F, dry1; C, G, dry2; D, H, dry3). Secondary xylem (A–D): tree-ring boundaries (inclined arrows) and narrower vessels communicating with wider vessels (horizontal arrows). Secondary phloem (F–H): discontinuous sclerenchyma bands (arrows). R, rhytidome; scl, sclerenchyma; SF, secondary phloem. Scale bars = 200 μm (A–D); 400 μm (E–H).

After each analysis, we visually inspected the residuals to verify the assumption of normality and homogeneity of variances. Tukey's post-hoc test ($p < 0.05$) was carried out to compare the populations if we detected a significant relationship in previous models. All statistical tests were performed in R v. 3.1.3 (R Core Team 2015), using the packages *FactoMineR* (Husson *et al.* 2016), *lsr* (Navarro 2015), *MASS* (Venables & Ripley 2002), *multcomp* (Torsten *et al.* 2008), and *vegan* (Oksanen *et al.* 2019).

RESULTS

Qualitative variation of the vascular system of stems across populations

The qualitative descriptors of the secondary xylem did not vary across populations in *M. polymorphum* (Fig. 2A–D; Appendix F at 10.6084/m9.figshare.16903063), and one qualitative descriptor of the secondary phloem varied across populations. In this case, the variation occurred in the sclerenchyma arrangement, in which discontinuous tangential bands of sclerenchyma were found exclusively in populations from the dry3 and wet sites (Fig. 2E–H; Appendix F at 10.6084/m9.figshare.16903063).

In populations of *Z. rhoifolium*, considering all 40 qualitative descriptors of secondary xylem, four varied across populations: gelatinous fibers, intercellular canals of traumatic origin, prismatic crystals, and pith flecks, which were observed only in trees from drier sites. Gelatinous fibers were found in dry3 and dry2 (Fig. F5A in Appendix F at 10.6084/m9.figshare.16903063). Intercellular canals of traumatic origin (Fig. 3D; Fig. F5E in Appendix F at 10.6084/m9.figshare.16903063) were observed at all dry sites. Prismatic crystals in chambered axial parenchyma cells were observed next to the intercellular canals of traumatic origin (Fig. F5E in Appendix F at 10.6084/m9.figshare.16903063) only in dry1. Pith flecks (Fig. 3B) were found at dry1 and

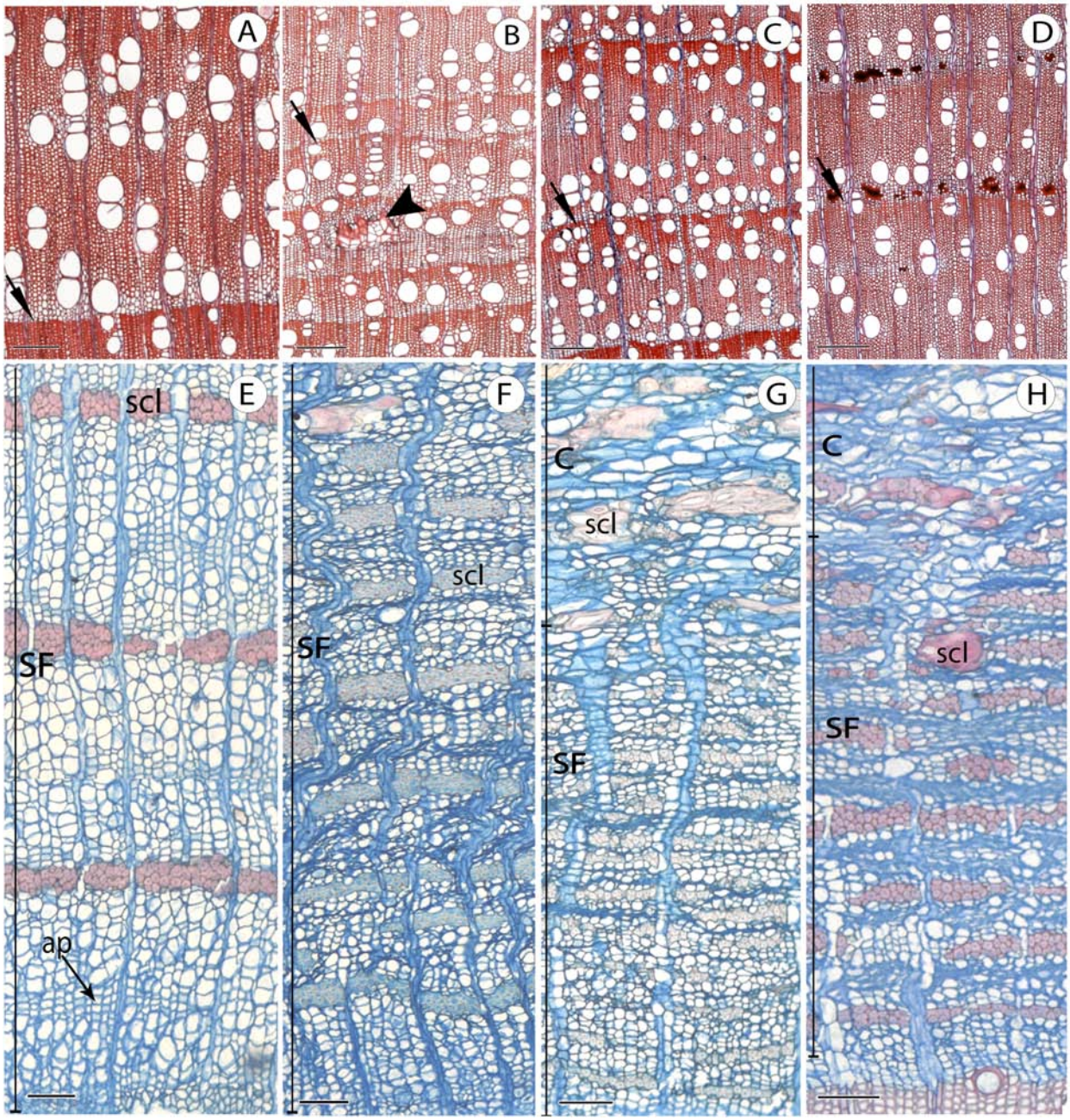


Figure 3. Cross-population variation of the secondary vascular system anatomy in stems of *Zanthoxylum rhoifolium* in transverse section (A, E, wet; B, F, dry1; C, G, dry2; D, H, dry3). Secondary xylem (A-D): tree-ring boundaries (inclined arrows); pith flecks (arrowhead). Secondary phloem (F-H): discontinuous tangential bands of axial parenchyma (ap). C, cortex; scl, sclerenchyma; SF, secondary phloem. Scale bars = 200 µm (A–D); 100 µm (E–H).

dry3. For the secondary phloem of stems of *Z. rhoifolium* (Fig. 3E–H), one qualitative descriptor varied across populations, the arrangement of the axial parenchyma in discontinuous tangential bands were observed only at the wet site (Fig. 3E). The qualitative description of the secondary xylem and phloem of *M. polymorphum* and *Z. rhoifolium* is detailed in Appendix F at 10.6084/m9.figshare.16903063.

Quantitative variation of the vascular system of stems across populations

In *M. polymorphum*, the first three PCA axes explained 59% of the total variation of the quantitative descriptors of the vascular system; the first axis explained 27% (λ = 4.61), the second axis explained 20% (λ = 3.44; Fig. 4A, Table 2), and the

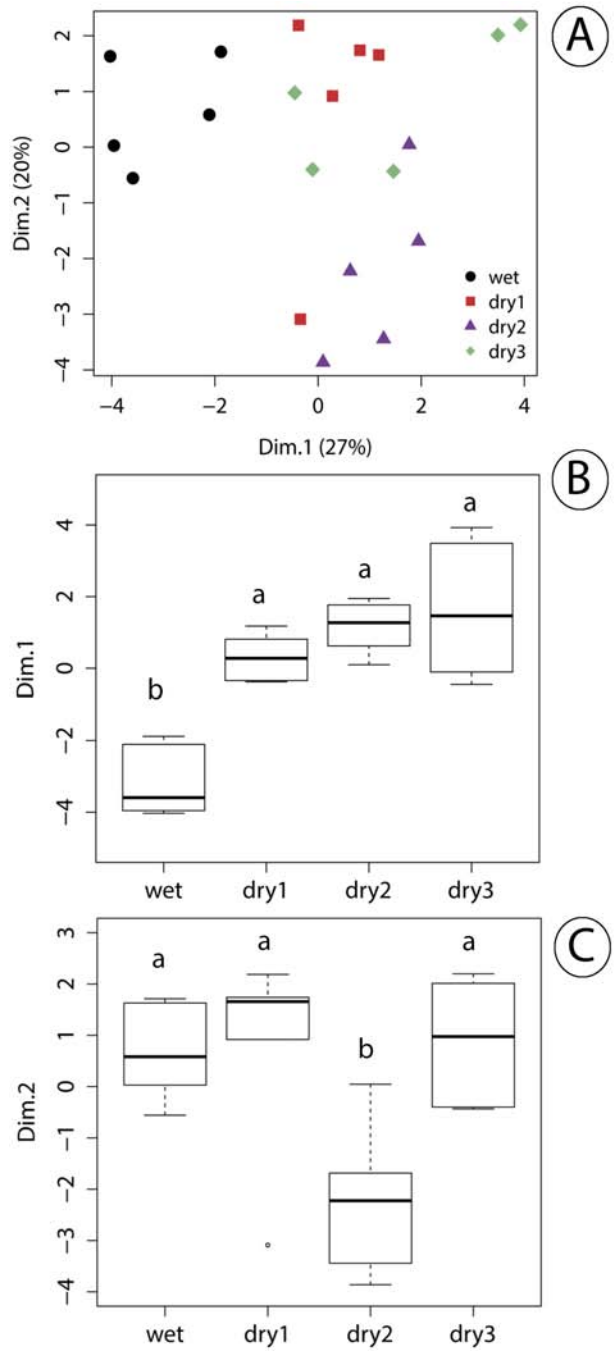


Figure 4. Cross-population variation of the secondary vascular system in stems of *Moquiniastrum polymorphum*. The first graph illustrates the first two PCA axes describing the variation of the secondary vascular system between plants from different populations (A). Differences along the first PCA axis among populations (B), and differences along the second PCA axis among populations (C). Different letters indicate differences among plant populations in post-hoc Tukey test ($p < 0.05$). In (B) and (C), boxes represent interquartile range, horizontal line inside the boxes represents the median, whiskers represent the 1.5*interquartile range, and dots represent points outside the 1.5*interquartile range.

third axis explained 12% ($\lambda = 2.04$) of this variation (Appendix G at 10.6084/m9.figshare.1690306З). As the anatomical variables were correlated only with the first two axes (Appendix G at 10.6084/m9.figshare.1690306З), we used both axes to test differences across sites. The vessel density, fiber wall thickness of the xylem, and ray height of both xylem and phloem were strongly correlated with the first PCA axis (Table 2). The lengths of the vessel elements and sieve-tube elements were strongly correlated with the second PCA axis (Table 2).

Table 2.
Pearson correlation coefficients (r) in pairwise comparisons of the quantitative descriptors of the vascular system and the first two PCA axes of the two tree species.

	Descriptor	*Moquiniastrum polymorphum*		*Zanthoxylum rhoifolium*	
		PC1 (27%)	PC2 (20%)	PC1 (25%)	PC2 (21%)
Secondary xylem	Vessel diameter	−0.62	−0.36	−0.71*	0.56
	Vessel density	0.81*	0.12	0.41	−0.29
	Vessel element length	−0.02	0.85*	0.71*	0.60
	Vessel grouping index	−0.10	0.48	−0.33	0.16
	Fiber-wall thickness	−0.86*	−0.18	0.00	−0.29
	Fiber length	−0.33	−0.01	0.14	0.65
	Rays per millimeter	−0.11	0.37	−0.54	0.72*
	Ray height	−0.81*	0.03	0.58	0.37
	Ray width	−0.62	−0.61	0.02	−0.67
	Intervessel pit aperture	−0.56	0.25	–	–
	Vessel-ray pit aperture	−0.62	0.54	−0.75*	−0.17
	Fraction of parenchyma	0.04	−0.65	−0.34	−0.15
Secondary phloem	Sieve-tube diameter	−0.22	0.49	−0.53	0.31
	Sieve-tube element length	−0.05	0.74*	0.50	0.57
	Fiber-wall thickness	–	–	−0.70*	0.00
	Rays per millimeter	0.16	−0.37	–	–
	Ray height	−0.79*	−0.14	0.56	0.24
	Fraction of cells with primary walls	0.52	−0.28	−0.31	0.63

The percentage of variance explained by each PCA axis is shown in parentheses. Some descriptors were not measured in either species and are not included in the table (see Methods).

* Correlation coefficients significantly different from zero ($p < 0.05$).

Table 3.
PCA axis variation explained by populations (sites) and plant height.

	Moquiniastrum polymorphum				*Zanthoxylum rhoifolium*			
	R^2	Fixed effects	p-value	n	R^2	Fixed effects	p-value	n
PCA1	0.71	Population	<0.001*	20	0.48	Population	0.003*	20
		Plant height	0.14	20		Plant height	0.52	20
PCA2	0.34	Population	0.02*	20	0.66	Population	0.0001*	20
		Plant height	0.46	20		Plant height	0.43	20

* Statistically significant ($p < 0.05$).

The first two PCA axes were explained by populations but not by plant height in *M. polymorphum*. The population from the wet site had the lowest values of the first PCA axis compared to all drier sites ($F_{3,19} = 14.96$; $p < 0.001$; Fig. 4B). Therefore, considering the anatomical descriptors strongly correlated with the first PCA axis (Table 2), high vessel density, thicker xylem fiber walls, and taller rays of both xylem and phloem tissues were observed in populations from the drier sites. Regarding the second PCA axis, the population from dry2 had the lowest values of the second PCA axis ($F_{3,19} = 4.61$; $p = 0.02$; Fig. 4C). As the conducting cells were strongly correlated with the second PCA axis (Table 2), shorter vessel and sieve tubes elements were in populations from dry2. The tree height in *M. polymorphum* did not explain the first ($F_{3,19} = 12.8$; $p < 0.001$) and second ($F_{3,19} = 3.5$; $p = 0.03$) axes of PCA (Table 3).

In *Z. rhoifolium*, the first four PCA axes explained 70% of the total variation of the quantitative descriptors of the vascular system, with the first axis explaining 25% ($\lambda = 4.03$), the second axis explaining 21% ($\lambda = 3.32$; Fig. 5A; Table 2), the third axis explaining 14% ($\lambda = 2.31$), and the fourth axis explaining 10% ($\lambda = 1.62$) of this variation (Appendix G at 10.6084/m9.figshare.16903063). As the anatomical variables were correlated only with the first two axes (Appendix G at 10.6084/m9.figshare.16903063), we used both axes to test differences across sites. The vessel diameter, vessel element length, vessel-ray pit aperture, and phloem fiber wall thickness strongly correlated with the first PCA axis (Table 2). The number of rays of xylem per millimeter strongly correlated with the second PCA axis (Table 2).

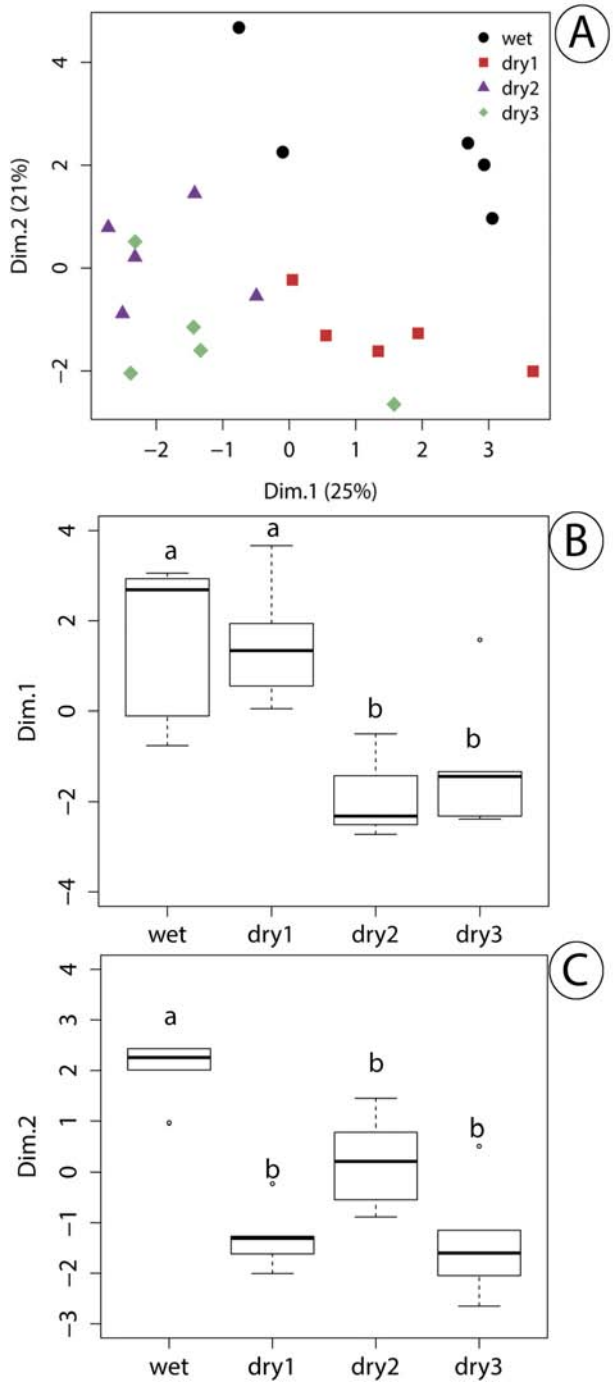


Figure 5. Cross-population variation of the secondary vascular system in stems of *Zanthoxylum rhoifolium*. The first graph illustrates the first two PCA axes describing the variation of the secondary vascular system between plants from different populations (A). Differences along the first PCA axis among populations (B), and differences along the second PCA axis among populations (C). Different letters indicate differences among plant populations in post-hoc Tukey test ($p < 0.05$). In (B) and (C), boxes represent the interquartile range, horizontal line inside the boxes represents the median, whiskers represent the 1.5*interquartile range, and dots represent points outside the 1.5*interquartile range.

The first two PCA axes in *Z. rhoifolium* were explained by populations but not by plant height. Populations at the wet and dry1 had the high values of the first PCA axis in comparison with dry2 and dry3 ($F_{3,19} = 7.33$; $p = 0.002$; Fig. 5B). Therefore, considering the descriptors strongly correlated with the first PCA axis (Table 2), the wet and dry1 populations had wider vessels, longer vessel elements, larger vessel-ray pit aperture, and thicker fiber walls in the phloem. The population at the wet site had the highest value on the second PCA axis ($F_{3,19} = 13.98$; $p < 0.001$; Fig. 5C). Because the number of rays in the xylem

Table 4.
Relationship between the phloem quantitative descriptors and xylem quantitative descriptors, highlighting the percentage of variation explained by the xylem in the models.

Response variable	*Moquiniastrum polymorphum*			*Zanthoxylum rhoifolium*		
	PCA 1 and 2 (xylem)			PCA 1 and 2 (xylem)		
	R^2	*p*-value	*n*	R^2	*p*-value	*n*
PCA1 (phloem)	0.09	0.17	20	−0.02	0.46	20
PCA2 (phloem)	−0.06	0.63	20	0.06	0.14	20

The quantitative descriptors of both tissues are represented by the first and second axes of the PCA.

was strongly correlated with the second PCA axis (Table 2), more numerous rays were in the xylem of the population from the wet site. The tree height in *Z. rhoifolium* did not explain the first ($F_{3,19}$ = 5.4; p < 0.006) and second ($F_{3,19}$ = 10.4; p = 0.003) axes of PCA (Table 3).

Mirrored pattern of xylem and phloem cells

In both species, we found simple plates in the conducting elements; the same cellular composition of the rays, with the ray body composed of procumbent cells and one or two marginal rows of square or erect cells; and the same number of cells (2-4 cells) in the axial parenchyma strands (Appendix F at 10.6084/m9.figshare.16903063). *M. polymorphum* had perforated ray cells in the xylem (Fig. F2G in Appendix F at 10.6084/m9.figshare.16903063) and ray sieve-tube elements in the phloem (Fig. F4C in Appendix F at 10.6084/m9.figshare.16903063).

The xylem quantitative descriptors and the tree height did not explain the variation of the quantitative descriptors of the phloem quantitative descriptors represented by the first ($F_{3,19}$ = 1.93; p = 0.17) and second ($F_{3,19}$ = 0.47; p = 0.63) PCA axes in *M. polymorphum*; and represented by the first ($F_{3,19}$ = 0.82; p = 0.45) and second ($F_{3,19}$ = 1.44; p = 0.26) PCA axes in *Z. rhoifolium* (Table 4).

DISCUSSION

We expected, for the first question, that the xylem and phloem structure of stems would differ across populations in both species. Indeed, we found cross-population differences in stem structure of the secondary xylem and phloem in both tree species, although the responses of the two species were different. For the populations of *Moquiniastrum polymorphum*, we found higher vessel density, thicker xylem-fiber walls, and taller rays in the xylem and phloem in trees from all dry sites, which agreed with our expectations. Also, for the populations of *Zanthoxylum rhoifolium*, we found larger vessels and vessel-ray pits, and longer vessels elements in individuals from the wet site, which agreed with our expectations, although one of the dry sites had the same features. In addition, for the population of *Z. rhoifolium* from the wet site, we found thicker-walled phloem fibers and more numerous rays in the xylem, which have not agreed with our expectations. Although the results were generally in agreement with our original expectations following the variation in water availability, there were specific and important differences in the secondary vascular system as discussed below.

Answering our second question, that the anatomical features of the xylem and phloem would have a mirrored structure in both plant species, we found similar character states of the perforation and sieve plates, the cellular composition of the rays, and the number of cells in the axial parenchyma strands were observed for both species. However, the quantitative descriptors of cell size were not correlated between the xylem and phloem tissues. Therefore, we have not observed a xylem-phloem mirrored structure of the conducting, support, and storage cells sizes.

How does the xylem and phloem structure vary across populations under different conditions of water availability?

Some qualitative descriptors varied across populations from the wet and dry sites, especially in *Z. rhoifolium*, and canals of traumatic origin and pith flecks were observed only in the xylem of trees from the dry sites. Because both structures are formed when wounds occur in the cambium (Carlquist 2001; Schweingruber 2007), such in response to low water availability (Schweingruber 2007), we suggest that, at these sites, these populations are more susceptible to occasional damage. Together with canals of traumatic origin, we also found prismatic crystals in trees from one of the dry sites (dry1). Environmental factors may be responsible for the abundance of crystals in the plant body (Franceschi & Nakata 2005) and can also be related to lower water availability, as observed for other tissues (Brown *et al.* 2013).

One issue related to the qualitative descriptors is the presence of distinct tree rings in both species at all sites. The ring presence across populations confirms that tree rings are not exclusively observed in species from drier or seasonal regions in the tropics, as previously detected (Marcati *et al.* 2006; Lisi *et al.* 2008; Blagitz *et al.* 2019) but are also observed in trees from wet sites. Thus, the growth rings in the xylem of *M. polymorphum* and *Z. rhoifolium* are genetically fixed, as they were present and had the same markers from sites under distinct hydrological conditions. The growth rings can be used to identify the species across populations, as previously reported (Mainieri *et al.* 1983; Sonsin *et al.* 2014).

The populations of *M. polymorphum* grown at the dry sites had vessel features associated with the maintenance of safer water transport. Higher vessel density in populations from dry sites increases the probability of the vessels to remain conducting if embolism occurs in adjacent vessels (Tyree & Zimmermann 2002). In general, a higher vessel density combined with reduced vessel diameters constitute a well-known relationship associated with safe transport (Sperry *et al.* 2008; Pratt & Jacobsen 2017). Although we did not fully detect it, remarkable features of this species in all populations can compensate for the lack of this relationship. *M. polymorphum* has a wide variation in vessel size (Fig. F1 in Appendix F at 10.6084/m9.figshare.16903063), which would prevent any trend of vessel diameter related to the environment being detected. Also, wider vessels are connected by narrower vessels (Fig. 2A–D), which can help to maintain the water flow through the vessel network (Brodersen *et al.* 2013). Connected vessels can provide alternative routes for water flow if adjacent vessels are embolized (Tyree *et al.* 1994; Carlquist 2012), improving the resistance to embolism spread (Mrad *et al.* 2021). Also, multiple terminal walls of the narrower vessels reduce the probability of air propagation between the wider vessels (Venturas *et al.* 2017). The helical thickening of the vessel wall in *M. polymorphum* (Fig. F1B, C in Appendix F at 10.6084/m9.figshare.16903063) creates an uneven inner-wall surface that increases the surface area of the vessels and therefore the area for water adhesion and consequently their wettability, which may help to remove small air bubbles (Kohonen & Helland 2009). The helical thickening also increases the mechanical strength of the wall and helps to conserve the vessel integrity even under high negative pressures (Jeje & Zimmermann 1979; Tyree & Zimmermann 2002). Thus, vessels of different widths connected to each other, and the helical thickening of their walls can help to assure water transport in the drier sites, compensating for the lack of reduced vessel diameter in this species.

Exclusively in individuals from site dry2, we observed shorter vessel elements that might be more resistant to collapse and deformation, due to the proximity of their walls (Sperry *et al.* 2006), playing a substantial and additional role in safer water transport in individuals from this site. Although we also observed shorter sieve tube elements in individuals from the site dry2, we cannot apply the same conclusion to the transport in phloem, because the analyses that included the length of the conducting cells in a phloem transport model based on the Hagen-Poiseuille equation (Thompson & Holbrook 2003b) concluded that the variation of the element length did not impact the phloem transport, which is more influenced by the sieve-tube diameter (Woodruff 2014).

Thicker-walled xylem fibers and taller rays of both xylem and phloem in populations of *M. polymorphum* can support safe water transport in the dry sites. The contribution of the fibers to the mechanical strength of vessel implosion has been noted, since the fibers that are in contact with the vessels can reinforce the vessel wall, reducing the possibility of vessel collapse under high negative pressures (Jacobsen *et al.* 2005). Thus, fibers with thicker walls may help to assure transport in trees from the dry sites. Taller rays in both the xylem and phloem at the dry sites can be advantageous during periods of lower water availability because the rays contribute to water storage (Borchert & Pockman 2005). Therefore, taller rays expand the volume for water storage in the secondary tissues at dry sites. Furthermore, as rays provide lateral communication between the secondary xylem and phloem (van Bel 1990; Sokolowska & Zagórska-Marek 2012), the water stored in the phloem can be moved to the xylem, supporting the water flow of this tissue by refilling the embolized vessels (Salleo *et al.* 2004). Thus, the phloem constitutes an additional water supply for the xylem, especially under dry conditions (Savage *et al.* 2016), and taller rays in the vascular system of *M. polymorphum* would optimize vessel refilling and prevention of embolisms.

Unexpectedly, in *Z. rhoifolium* populations, we observed similar anatomical features in dry1 and wet sites described by the first PCA axis, although not by the second PCA axis. This partial similarity in both sites could be caused by other environmental gradients not considered here. For example, these two sites are located at higher altitudes (Appendix A) that could condition plants with a higher root/shoot ratio (Leuschner *et al.* 2007). If so, populations of *Z. rhoifolium* from the wet and dry1 sites could access deep water at a higher altitude due to a greater plant investment in the root system. Although we do not have direct evidence of the root system in these plants, in some cases plants at higher altitudes can extract water from deeper soil layers due to site-specific traits of their root systems (Rossatto *et al.* 2012). Both populations of *Z. rhoifolium* had wider and longer vessel elements which can potentially conduct higher volumes of water (Tyree & Zimmermann 2002; Sperry *et al.* 2008; Hacke *et al.* 2017), and larger vessel-ray pit apertures that favor higher water flow rates between adjacent cells (Hacke & Sperry 2001). Thus, all these features can contribute to efficient water transport in these populations, although it partially contradicts our initial expectations for the populations of dry1. Also, plants of *Z. rhoifolium* from the wet and dry1 sites had

thicker-walled phloem fibers. The process of lignin deposition in the cell walls is a result of enzymatic activity, such as the peroxidases associated with the cell wall, which are induced by different factors, including a high concentration of heavy metals in the soil (Vangronsveld & Clijsters 1994). Therefore, the thicker-walled phloem fibers in both wet and dry1 populations might be associated with the higher aluminum contents in these environments (Appendix A at 10.6084/m9.figshare.16903063), since high aluminum contents were able to induce peroxidases involved in the lignification process (Jouili *et al.* 2010). Although the variation of lignified tissue features in the phloem has already been associated with the presence of mineral ion content in the soil (Schweingruber 2006), further research considering experimental evidence is needed.

Z. rhoifolium had more numerous xylem rays in the wet site, contradicting our expectations. Considering the lateral communication between xylem and phloem (van Bel 1990; Sokolowska & Zagórska-Marek 2012), more numerous rays would promote higher exchange rates of resources between these two tissues also in mesic environments, not being an exclusive feature of plants from dry environments. Furthermore, if we also consider that xylem rays can contribute to water and carbon storage (Słupianek e*t al.* 2021), more numerous rays can promote greater storage of these resources in the wet site population. This finding suggests the number of xylem rays may be less variable according to climate (see Morris *et al.* 2016) and can be related to other features that are intrinsic of individuals, such as tree leaf area (Gartner *et al.* 2000), although further research is necessary to better understand these relationships.

It is remarkable that populations of the two tree species had different features of the vascular system in sites with similar water availability. *M. polymorphum* had features that can promote and support safer water transport at each specific site, generally following our expectations. In contrast, *Z. rhoifolium* had vascular system features in disagreement with our expectations. In this case, we must consider that other environmental gradients beyond the water availability and species-specific constraints could occur, and *Z. rhoifolium* may be more impacted by them. We have not evaluated the whole-plant level, and morphological or structural variations of leaves, roots, and/or stem outer bark may be limiting changes in the wood traits investigated here. Another remark that should be considered is the high variation of the anatomical descriptors among trees within each population of *Z. rhoifolium* (Appendix H at 10.6084/m9.figshare.16903063), which could be concealing the effect of water availability at a larger scale in this environment. Thus, these findings highlight that species generalizations regarding the structure of the vascular system in dry and wet sites should be avoided. Regardless, future experimental studies in both species should evaluate the impact of water availability on different vascular tissues and cell types, and also on their phenotypic plasticity.

Is there a mirrored structural anatomy of the conducting, support, and reserve cells of the xylem and phloem tissues, regardless of the environmental conditions?

These species share isomorphic features, both axially and radially, highlighting the influence of the same origin of the cells (initials of the cambium). The rays showed more parity between the two species since their cellular composition was the same, and this structural and functional coordination can optimize the interaction of the two tissues (Vergílio *et al.* 2017), facilitating the resource balance between them.

Regarding the quantitative descriptors, the different selective forces for the specialization of each tissue had a stronger effect on the cell sizes than the ontogenetic origin of initials of the cambium. For example, the distinct hormonal balance between tissues and the auxin gradient further from the vascular cambium in both tissues (Bhalerao & Fischer 2014); the earlier differentiation of phloem than xylem (Montwé *et al.* 2019); the transverse or some other divisions that occur only in the sieve tube element mother cells (Khan & Siddiqui 2007); and the different mechanisms that regulate the behavior of intrusive growth, and consequently the elongation, between xylem and phloem fibers (Majda *et al.* 2021). Therefore, although many genes are activated concurrently at the time of vascular tissue initiation, each tissue specializes in a specific way from the moment they initiate their differentiation process (de Rybel *et al.* 2016), not resulting in a correlated cell size between the xylem and phloem tissues, resulting in non-mirrored structural anatomy related to the quantitative features of the vascular system.

CONCLUSIONS

Our study revealed specific differences between the structural descriptors of the vascular system and the water availability across populations at different sites. At dry sites, *M. polymorphum* populations had higher vessel density, thicker xylem fibers cell walls, and taller rays in both xylem and phloem, which can promote and support safe transport. In contrast, *Z. rhoifolium* populations had vascular system features that were disparate to the water availability in each site. Comparison of the xylem and phloem revealed mirrored axial and radial variations in these two tissues only for the qualitative features, mainly for the rays. Quantitatively, the xylem and phloem cells sizes were completely uncoupled, not mirroring each other. Thus, when

comparing secondary xylem and phloem, we found a dichotomy regarding the mirrored behavior between the qualitative and quantitative descriptors.

Finally, when we evaluated the variation of descriptors under contrasting conditions of water availability, we found a greater variation in xylem descriptors than phloem, indicating that xylem probably responds more to variations in water availability, while phloem has its performance without outside control in *M. polymoprhum*. In *Z. rhoifoilum*, both xylem and phloem performances seem to be decoupled to environmental water availability. The autonomy of development of each tissue in the two species can be related to the absence of a mirrored structure in the cell sizes of each tissue.

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